

Othmane Echchabi

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Machine learning researcher working on spatiotemporal modeling, remote sensing foundation models, and climate-risk forecasting, with applications in wildfire modeling, infrastructure monitoring, and environmental decision support.

EDUCATION

McGill University & Mila – Quebec AI Institute

M.Sc. in Computer Science (GPA: 4.0/4.0)

Montréal, QC

Expected 05/27

Duke University / Duke Kunshan University

B.S. in Data Science

Durham, NC / Suzhou, China

05/25

- Thesis: *Monitoring Spartina Alterniflora Using Self-Supervised Learning*.
- Study Abroad: Universidad Carlos III de Madrid (Spring 2022).

TECHNICAL SKILLS

ML & Systems : Python, PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, Docker, Kubernetes, Linux, Git, Java

Geospatial & Remote Sensing : Sentinel-2, satellite imagery, SSL foundation models, ArcGIS, geospatial inference, climate data

Research Areas : Spatiotemporal ML, computer vision, remote sensing, wildfire modeling, sustainability AI

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2025 – Present

Rolnick Lab, McGill University & Mila

Montréal, QC

- Collaborating with the OLMoEarth team at Ai2 (Allen Institute for AI) on wildfire foundation models, exploring remote-sensing and climate covariates for scalable fire-risk modeling and environmental monitoring.
- Leading a Smithsonian-collaborative tree-mortality classification project on Barro Colorado Island (BCI), Panama; developing georeferenced canopy-image datasets and computer-vision models for forest health assessment.

Machine Learning Research Intern

Jul. 2025 – Dec. 2025

College of Computing, University Mohammed VI Polytechnic (UM6P)

Benguerir, Morocco

- Integrated a 10-year crop-yield dataset with N/P/K fertilizer inputs across 150+ countries for downstream modeling.
- Built a transformer-based recommendation model and a RAG assistant for agronomy; reduced query latency to ~2s.

Undergraduate Research Assistant

Jun. 2024 – Aug. 2025

Sustainability Assessment Group & Chang Lab, Duke Kunshan University

Suzhou, China

- Built a 300k-image, ground-truthed satellite piped-water access dataset and achieved SotA accuracy with SSL foundation models; validated predictions against UN data by running large-scale inference across 50/54 African countries using 1M+ Sentinel images and population-density covariates.
- Built SpeedTransformer for Geolife transport-mode prediction (96% Top-1, +3% vs. baseline), and prototyped CarbonClever, a phone application, to deliver personalized guidance for reducing personal CO₂ emissions.

Undergraduate Research Assistant

Jan. 2025 – Jun. 2025

Malof Lab, University of Missouri

Columbia, MO

- Developed Perceiver IO-based models and a VAE simulator for wildfire spread forecasting from satellite, weather, and climate covariates; evaluated on WSTS+ to study scenario-driven fire progression and multimodal fusion strategies.

Undergraduate Research Fellow

Jun. 2023 – Aug. 2024

Data+ & Climate+, Duke University

Durham, NC

- Mapped invasive salt-marsh species from Sentinel-2 using ViTs; delivered landscape-scale outputs for conservation planning.
- Built a geospatial database on saltwater intrusion and sea-level rise; delivered an interactive ArcGIS dashboard.

PUBLICATIONS & PREPRINTS

1. Zhang, Y.*; **Echchabi, O.***; Feng, T.; Zhang, W.; Liao, H.-K.; Chang, C. *Detecting Transportation Mode Using Dense Smartphone GPS Trajectories and Transformer Models*. Accepted at *International Journal of Geographical Information Science* (2026).

2. **Echchabi, O.***; Lahlou, A.*; Talty, N.; Manto, J. M.; Tongshu, Z.; Lam, K. L. *Monitoring Access to Piped Water and Sanitation Infrastructure in Africa at Disaggregated Scales Using Satellite Imagery and Self-Supervised Learning*. Accepted at the *Machine Learning for Remote Sensing Workshop, International Conference on Learning Representations* (2026).
3. **Echchabi, O.***; Lahlou, A.; Talty, N.; Manto, J. M.; Lam, K. L. *Monitoring access to piped water and sanitation infrastructure in Africa at disaggregated scales using satellite imagery and self-supervised learning*. arXiv preprint.

*Equal contribution.

POSTERS & PRESENTATIONS

- Monitoring *Spartina alterniflora* Using Self-Supervised Learning. *Duke Climate+ Symposium* (2024).
- Assessing Climate Change Risk of Rural Coastal Plains. *Duke Climate+ Symposium* (2023).
- Lahlou, A.*; **Echchabi, O.***; Talty, N.; Manto, J. M.; Tongshu, Z.; Lam, K. L. *Scalable Water and Sanitation Monitoring Across Africa via AI-Driven Satellite and Survey Data Integration*. *American Geophysical Union Annual Meeting* (2025).

GRANTS & AWARDS

- Dan Ștefan Petrescu Fellowship – \$30,000 (2026)
- McGill CS Graduate Award — \$30,000/year (2025)
- McGill CS Graduate Registration Bonus — \$5,000 (2025)
- DKU Student Experiential Learning Fellowship — \$1,000 (2024)
- DKU Signature Work Research Grant — \$1,500 (2023)
- DKU Innovation Incubator — \$3,000 (2023)

TEACHING EXPERIENCE

- **McGill University** — Teaching Assistant (2025–Present) — COMP 330, COMP 370;
- **Duke Kunshan University** — Math & CS Teaching Assistant (2021–2025); Courses: COMPSI 101, COMPSI 201, MATH 105, MATH 201, MATH 202, MATH 317;
- **FADI Academy - Online College Prep Center** — Partner & Math Tutor

SERVICE & LEADERSHIP

- **Mila - AI Quebec Institute** — Elected Mila Lab Representative (1 of 11), representing 900+ students across Mila programs (2025–Present).
- **McGill University** — Project Development Vice President (Board Member), Muslims in Tech; coordinate technical programming for students and early-career members (2025–Present).
- **Duke Kunshan University** — Resident Assistant; Soccer Varsity Co-Captain; CS Club Software Tech Lead; organized DKU's largest hackathon and led technical workshops (2021–2025).

MISCELLANEOUS

Spoken Languages : Fluent in Arabic, French, English; Intermediate Spanish and Chinese

Interests : Competitive Soccer, Weightlifting, Cooking, Traveling.